MATH 190 Quiz 3

21 October 2026

- The quiz is 20 minutes long.
- The quiz consists of 4 pages and 2 exercises worth a total of 24 points.
- This is a closed-book examination. **None of the following are allowed**: notes, calculators, electronic devices, or anything other than pen.
- No work on this page will be marked.
- Show all of your work for full credit.
- Fill in the information below before turning to the questions.

Student number								
Section								
Name	. . . . .							
Signature								

Please do not write on this page — it will not be marked.

1. [12 marks] Let $f(x) = \frac{x-1}{x}$.

    1. Please use the definition of the derivative to compute $f'(x)$.
    2. Please check your answer by computing $f'(x)$ using any other method.

2. [12 marks] Please place your answer in the box provided. Compute the derivative of the following function using any method.

$$3x^5 - \sin(x) - xe^x + 2$$

Answer: